

Priscila Madrid

+1 (915) 706-6424 | madridpriscila2002@gmail.com | linkedin.com/in/priscila-madrid | github.com/prismadrid2002 | priscilamadrid.com

EDUCATION

The University of Texas at El Paso (UTEP)

B.S. Computer Science, Minor in Mathematics, GPA 3.80 / 4.00

El Paso, TX

Graduated May 2025

Honors: UTEP Top Ten Senior (2025), NCWIT Collegiate Award Finalist (2025), SHPE STEM Poster Winner (2024), Great Minds in STEM Scholar (2023, 2024), Grace Hopper Scholar (2023)

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, JavaScript, TypeScript, Bash, Java (coursework)
- **Frameworks & Libraries:** pandas, NumPy, matplotlib, PyTorch, Next.js, React
- **Tools & Platforms:** Git, GitHub, Node.js, Firebase, Claude API, Appium, WinAppDriver, Jira, Zephyr, IBM SPSS Statistics, Microsoft Excel, Linux, HPC clusters

EXPERIENCE

UTEP Institute for Applied AI Innovation (AAIL)

Research Assistant

El Paso, TX

October 2025 to Present

- Ran the AI Connect pilot analysis on 12 surveys in **IBM SPSS Statistics** and **Excel** (cleaning, validation, restructuring, **nonparametric tests**) to prepare the methodology and codebook for the 500 survey study now in collection.
- Prototype, in AAIL's translational lab, a local **Claude API** proxy in **Python** and **TypeScript** with **REST** endpoints that route requests across Claude models by task and complexity, log usage for cost tracking, and handle errors.
- Support a \$180,000 AI consulting engagement over seven months with a regional manufacturer: lead stakeholder discovery and write the **Software Requirements Specification (SRS)** for three AI pilots and a readiness assessment.
- Document, for the translational lab, a proposed secure deployment architecture for UTEP AI applications: **GitLab**, dev / test / prod VMs, security scanning, firewalls, Research and Academic Data Center storage, and HPC / GPU compute.

MIT Department of Earth, Atmospheric, and Planetary Sciences

Summer Research Intern

Cambridge, MA

June 2024 to August 2024

- Trained **PyTorch CNNs** on the **Bridges-2** supercomputer to parameterize subgrid convection in climate models.
- Tested **CNN** architectures (three convolutional layers, ReLU, dropout) as candidates to replace a fully connected network, and benchmarked training runtime.

W. M. Keck Center for 3D Innovation

Undergraduate Research Assistant

El Paso, TX

January 2024 to May 2024

- Decoded the .ILT and .CLI printer file formats and built a **Python** tool (**pandas**, **NumPy**, **matplotlib**) that transformed a CSV **design of experiments** (5 factors, 3 levels, 243 runs) into 243 **AconityMIDI+** build files.
- Cut a manual configuration workflow of about two weeks to about 54 seconds by automating generation.

Eaton, Thomas A. Edison Technical Center

Software Quality Assurance Intern

Franksville, WI

May 2023 to August 2023

- Built a **Python** test automation framework with **Appium** and **WinAppDriver** for UI testing of **ProView NXG** (Form 7 control software), intended to replace licensed tooling (about \$30,000 in licenses plus \$15,000 in annual maintenance).
- Helped design the **Falcon Continuous Integration and Continuous Testing** pipeline: a watcher launched by a timer or an external trigger to install builds, run test suites, and publish results to **Zephyr**, **Jira**, and **GitHub**.

SIDE PROJECTS

UTEP Data Diggers, Faculty Research Data Management System

SQL Lead, Databases course team project

El Paso, TX

2024

- Designed ER diagrams, the relational model, and a **normalized schema** for students, faculty, projects, and supervisors, and implemented the schema in **MySQL** with **primary keys**, **foreign keys**, and **unique constraints**.
- Translated 12 functional requirements into the data model and presented scope, ER diagrams, and relational model.

Bloomberg Tech Lab on Campus, MarketWatch RabbitMQ Lab

Lab Fellow

El Paso, TX

2025

- One of 40 students selected to collaborate with Bloomberg engineers: built producer and consumer services in **Python** with **pika**, publishing and consuming securities price updates through a **RabbitMQ** message broker in **Docker**.
- Implemented a **topic exchange** that streams stock price updates to queues by ticker and sector using wildcard binding keys for selective subscriptions.

LEADERSHIP

UTEP Computer Science and Computing Alliance of Hispanic-Serving Institutions (CAHSI)

CS Engagement and CAHSI Backbone Assistant

El Paso, TX

January 2025 to October 2025

- Supported 3 clubs and 8 student organizations and coordinated logistics for visits from large technology companies.
- Organized with **CAHSI** the **Great Minds in STEM (GMIS)** conference delegation and its student travel logistics.